

NexusNTU

Release Plan

Version 1.0 approved

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1 Introduction

This document details the release plan specifications and scope of the NexusNTU project, an all-in-one digital solution designed for international students at Nanyang Technological University (NTU).

1.1 Purpose

This document shall lay out the release plan for NexusNTU, which is a Progressive Web Application (PWA) designed as an all-in-one digital companion for international students at NTU, along with the relevant supporting documentation needed. The purpose of the release plan serves to manage, plan, schedule, and control the application build through different stages and environments, including testing and deploying software releases. This also helps create an actionable plan built on specific features, enhancements, and fixes to build, ensuring alignment with the project timeline and institutional requirements.

1.2 Scope

This document consists of the release strategy of NexusNTU and will provide relevant details for all past, present, and future releases of the application and documents. This document shall be updated whenever there is a new build or release by the developmental team or changes made to the various documents by the QA team, which will allow us to record and keep track of all releases of the application and documents throughout the lifecycle of the project. The release engineer shall work closely with the project manager, developmental team, and QA team to plan release windows and cycles and document all version changes.

1.3 Intended Audience

The intended audience of NexusNTU is primarily the population of international students at Nanyang Technological University. They are the target demographic experiencing the current fragmented and inefficient onboarding processes. The document also serves the developmental team and the quality assurance team, providing a roadmap for product functionality and testing requirements.

1.4 Expected Evolution

The expected evolution of the platform will include continued maintenance to ensure the system is up-to-date with new technology advances, along with fixing bugs and adding new functionalities based on continuous user feedback. Future expansion of the scope is anticipated, potentially including the development of native applications (currently constrained to PWA) or pursuing deeper API integrations should access to proprietary NTU systems become available.

1.5 Security and Privacy Considerations

To maintain user privacy, the User Account and Authentication feature requires secure log-in procedures, including two-factor authentication. Crucially, the application is designed only to store non-sensitive metadata and user preferences. The system will not handle or store sensitive personal data, such as passport numbers or FINs.

2 Referenced Documents

Document Name	Document Number	Issuance Date
System Requirement Specifications	SRS	23 September 2025
Quality Plan	QP	2 October 2025
Risk Management Plan	RMP	14 October 2025
Project Plan	PP	14 October 2025

3 Overview

NexusNTU is designed as an all-in-one Progressive Web Application (PWA) intended to be a comprehensive digital companion for international students at Nanyang Technological University (NTU). This application comprises several main features: a Centralised Information Hub, Campus Navigation, an Amenities Finder, and an AI-Powered Support Chatbot. The core functionality aims to solve the problem of a fragmented and inefficient onboarding process by centralizing essential information and services into a single, user-friendly platform, thereby helping students integrate smoothly into university life. The NexusNTU platform targets the common challenges international students face when navigating multiple disconnected university websites, physical offices, and third-party systems, a fragmentation that typically leads to information overload, confusion, and delays during their critical transition period.

The project utilizes the Model-View-Controller (MVC) Architecture 1 due to the need for dynamic changes to the view based on evolving feature sets. This architectural choice facilitates the separation of concerns, allowing for independent development and modification of data (Model), user interface (View), and application logic (Controller). The initial scope includes developing the application as a PWA, accessible across mobile and desktop browsers, rather than native iOS or Android apps. It relies on deep-linking to official NTU systems for integration, as proprietary back-end access is unavailable, and the system will only store non-sensitive metadata and user preferences.

The development follows an Agile methodology with incremental design and implementation, ensuring the product is delivered without error. The initial design likely began with an internal skeleton (similar to Version 0.1 of other projects) focused on foundational architecture. Subsequent versions will integrate and refine the core features, such as secure User Account and Authentication (including two-factor authentication), the Campus Navigation map, the Amenities Finder, and the 24/7 AI-Powered Chatbot. The final release will be the culmination of iterative testing and refinement, ensuring a high-quality, comprehensive digital companion for international students.

The use case diagram that provides a high-level context of the system is:

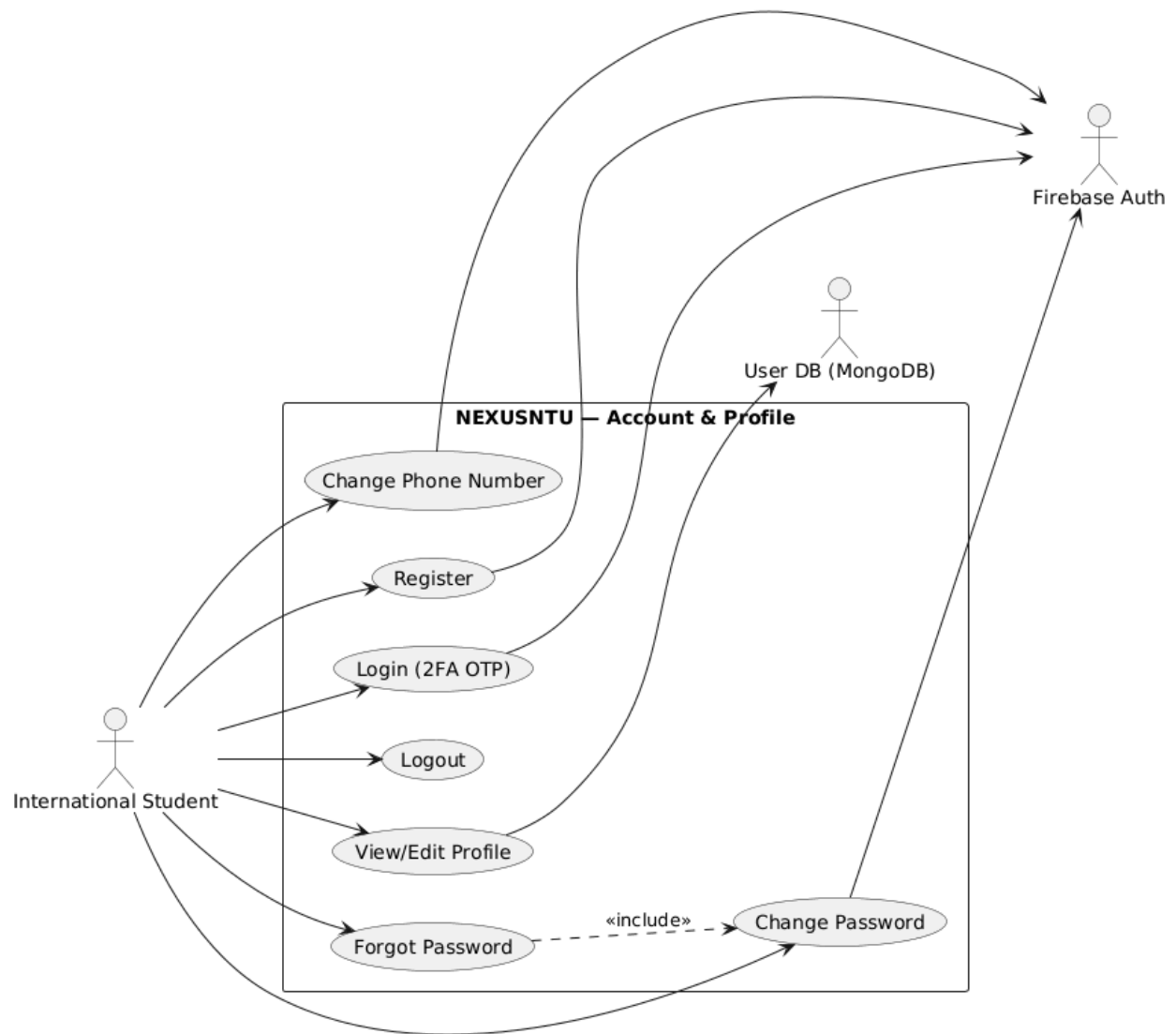


Figure 1: Use Case Diagram

4 Assumptions, Constraints, Risks

4.1 Assumptions

This section identifies the statements believed to be true for the successful release of the product and completion of the project

Technology

- The main framework of the system (Progressive Web Application base and core architectural pattern) is to remain unchanged after release.
- Required external dependencies, such as NTU's official data feeds (for navigation and amenities) and the third-party AI service provider (for the Support Chatbot), are assumed to continue to be functional and available after the application's release.
- It is assumed that the PWA framework adopted for the front-end will maintain cross-platform compatibility across major mobile and desktop operating systems.

Budget and Funding

- The estimation of cost and resource allocation is accurate, and no sudden or additional large amount of cost is incurred.
- Sufficient institutional support or project funding is secured for the acquisition of necessary development tools and external API licenses (if required) on time for development.

Development

- NexusNTU's developers are appropriately skilled and able to meet expectations and the timeline of the project development lifecycle.

Marketing

- NexusNTU requires official promotion and endorsement from the NTU International Office or Student Affairs department for the application to be well-known and widely adopted by the target audience, the international students.

4.2 Constraints

This section identifies any limitations that must be taken into consideration prior to the release of the product.

Resources

Users' data on the application, including usage metrics and stored preferences, are still owned by the users. All rights of using data must be clearly given prior approval by the users before being used or stored in an external database or software.

All data handling must strictly adhere to the university's policies on student data privacy and confidentiality (PDPA compliance).

Time

The entire project from the planning to the release stage aims to be completed within the stipulated academic timeline. To ensure no delays, project planning must be cautious and correctly estimated

Budget

With limited budget, proper planning and budgeting must be carried out to ensure processes do not cause a budget overrun.

Manpower

The project team size is small. Careful work assignment and task planning must be carried out, and the project manager must exercise a structured and flexible manpower allocation to ensure project tasks can be completed on time.

Functionality

The Campus Navigation and Amenities Finder functionalities are dependent on the accuracy and accessibility of NTU's mapping and database systems. Proper offline functionality must be maintained for core information display to ensure students can access essential data even if connectivity is intermittent. The AI-Powered Support Chatbot functionality is dependent on a connection to the external AI service provider API and the user's internet connection.

4.3 Risks

Risk	Impact Severity	Probability	Mitigation Strategy
Unable to Meet Project Deadline	High	Medium	Follow Agile methodology with sprints to track progress closely. Have regular meetings to identify and resolve blockers early.
Dependency on External APIs	High	Medium	Develop fallback mechanisms (e.g., show cached data, deep links) for when an API is down. Abstract API integrations to allow for easier replacement if needed.
Feature Creep	High	Medium	Strictly adhere to the defined scope in the SRS. All new feature requests must go through a formal change request process and be approved by the Project Manager.
Team Member Unavailability	Medium	Medium	Maintain clear and updated documentation so other team members can take over tasks. Foster knowledge sharing within the team.
System Not	High	Low	Conduct User Acceptance Testing (UAT) with the

Meeting Requirements			prototype early in the process. Maintain a requirements traceability matrix to ensure all specifications are met.
Data Security Breach	High	Low	Use a trusted authentication service (Firebase). Avoid storing any sensitive PII. Follow security best practices for web development (e.g., input sanitization, secure API keys).
Broken Deep Links	Medium	High	Implement a link-checking mechanism or a user-reporting feature for broken links. Admins will regularly verify and update the links in the database.

5 Release Approach

5.1 Rationale

As mentioned in the Project Plan document and earlier in this document, the Agile methodology is used to develop the NexusNTU application. The main idea of Agile is to develop the program with small incremental releases in iterations by breaking down a bigger problem into multiple sub-problems. Changes, such as the implementation of new features, can be made in an incremental manner and delivered through collaboration between self-organizing cross-functional teams.

Agile also provides a constant feedback mechanism while the product is delivered. Additionally, the Agile approach testing is integrated with the project lifecycle which begins very early in the software development process. The increased frequent testing allows less effort to resolve the bugs later and a higher quality product delivered at the end of the NexusNTU project.

The Agile development approach is also selected due to the short time frame given for the project. This method will allow for changes to be easily made within the limited duration of the project, which is extremely crucial to meet and deliver the complete product by the deadline planned. Even when most requirements are well-defined at the initial stages of the project, addition and changes of functionality are not uncommon. Moreover, due to the independent nature of our individual functionalities (Campus Navigation, Amenities Finder, Support Chatbot, and Information Hub), it would be more efficient for us to split the development of the functionality for sprint planning. Hence, the Agile methodology helps us to complete the project within the time constraint specified in the previous section.

Therefore, the Agile development methodology is the most ideal methodology and chosen over the other methodologies to ensure that the releases delivered to users are reliable and satisfy requirements. Compared to the traditional waterfall methodology, the Agile approach is a much more flexible and appropriate model to follow.

5.2 Release Strategy

The release strategy will employ a phased function rollout. In a phased rollout, implementation of functions happens in stages. Information gleaned from earlier implementations can assist subsequent implementations so that potential issues can be minimized. This aligns well with the Agile methodology as mentioned in the above rationale as iterations are developed incrementally.

To facilitate the release process, continuous integration and the Build-Package-Deploy process will be used. Continuous integration involves merging individual components alongside

development, as opposed to merging many components at once. This is a good practice that allows all team members to keep up with the current latest build, reducing time spent on miscommunication and development conflicts. The Build-Package-Deploy process includes running unit tests, generating release notes, and running integration tests before distribution of the product.

Version control software like Git is also used to manage releases. Such software provides a convenient way to view the history of changes to the product as well as to revert to a previous version of the product if need be. Branches can also be created to allow team members to work in parallel—which reduces time spent—and merge their changes in the future.

Alpha and Beta testing will be used to verify that the product satisfies requirements without issues or bugs. Feedback gained from these tests will be used to make final improvements to the product before release. An automated approach is used for release distribution. Installation of the product will be handled through the university’s hosting infrastructure and accessible via a centralized web URL for cross-platform PWA access.

After alpha and beta testing, the first major release will be made. It will include some of the main functionalities of the application, which are specified in the System Requirements Specification document. Subsequent minor releases will be made based on feedback from users, which will include bug fixes and improvements in user experience. Subsequent major releases will include more main functionalities on top of the previous major release. Additionally, all releases will be internally tested by team members before they are distributed to users.

5.2.1 Release Content

The following table showcases what our team will release in each version of the software application:

Release Version	Release Content	Remarks
0.1	Skeleton of application with basic UI design and layout.	-
1.0	Bug fixes and improvement in UI.	-
1.1	Bug fixes and notification system.	-

5.2.2 Release Schedule

The Release Schedule is an important timeline that our team will strictly adhere to in order to maintain a consistent frequency of releases. This allows us to uphold user satisfaction through constant quality maintenance and regular updates of new features. The following table states the expected delivery dates for the different release versions after thorough discussions with our stakeholders and the university.

Release Version	Release Content
0.1	23rd October 2025
1.0	30th October 2025
1.1	7th November 2025
2.0	12th November 2025

5.2.3 Release Impacts

With every release, several components of our project are impacted. The following table summarizes how our business process and system are impacted with each release. We also state the benefits, goals and objectives that our project aims to achieve for every release.

Release Version	Impact on Business Process	System Impact	Benefits, Goals and Objectives
0.1	Initial UI screens with no functionality	Implementation of PWA frontend architecture	Provides a working interface that helps the team to understand the flow of navigation through the app
1.0	- Complete essential features of NexusNTU as a product - Announcement of product and rollout to international students - Setup of channel for receiving user feedback	- Main branch now ready for new changes - UI elements must be functional; elements may be changed to better suit the functionality specified - Requires establishment of backend database and proper encryption of user credentials - Implementation of main functionality according to functional requirements	- First major release to users - Feedback from users used for bug fixes and improvement - Provides centralized access to campus information and navigation

1.1	- Announcement of minor release to users - Improvement of product value through bug fixes	- Minor release branch tagged and release deployed to user - Improve reliability of the application	- Bug fixes and UI enhancements - Achieves a higher level of usability and clarity - Increase user satisfaction and adoption rates
2.0	- Further refine product value through new features - Announcement of major release to users	- Parallel update of main and release branches in Git version control - Requires integration with external AI service provider API - Implement notification system for critical alerts and updates	- Enhanced user support through AI-powered chatbot - Implement push notification capabilities for time-sensitive information - Increase engagement and improve student experience

5.2.4 Release Notifications

With every release, several components of our project are impacted. The following table summarizes how our business process and system are impacted with each release. We also state the benefits, goals and objectives that our project aims to achieve for every release.

Stakeholders	Method of Notification	Notification Information	Timeframe for Receipt of Notification
International Students (Users)	Email notifications, in-app announcements, and university communication channels (official website, student portal)	Changes to be made, including bug fixes, changes to UI, new features, and improvements to existing functionalities	7 days prior to release version for email notification and website update; in-app notifications will display upon next app access
Development Team	Email, instant messaging, and team meetings	Same information as provided to users; additional information regarding reasons for changes and any changes to documentation,	Immediately after the plan for a release is approved

		architecture, or technical specifications	
University Stakeholders (International Office, Student Affairs, IT Department)	Email and formal meetings	Release schedule, key features, system impacts, and any necessary institutional actions or communications	14 days prior to scheduled release for major versions; 7 days for minor updates

6 Glossary

The plan is that the items listed below will be delivered by the stated deadlines.

Term	Definition
Agile	A set of practices centered around the idea of iterative development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams.
AI-Powered Chatbot	A key feature within NexusNTU designed to provide instant, automated support and answers to common international student inquiries using artificial intelligence models.
Centralized Information Hub	A core component of the application that consolidates essential university information, documentation, and official links into a single, easily navigable location.
Git	A free and open-source distributed version control system used to manage the history of changes to the source code and facilitate parallel development via branching.
Model View Controller (MVC)	An architectural pattern that separates the application logic into three interconnected components: the Model (data management), the View (user interface), and the Controller (application logic).
NexusNTU	The name of the Progressive Web Application (PWA) being developed as a comprehensive digital companion for international students at NTU.
Progressive Web Application	A type of web application that utilizes modern web capabilities to deliver a cross-platform, app-like user experience accessible through standard web browsers.
Release	A defined version of the NexusNTU application made available for users via the university's designated web link.

7 Acronyms

Acronym	Definition
PWA	Progressive Web Application
MVC	Model-View-Controller (Design Pattern)
AI	Artificial Intelligence
SRS	System Requirements Specification (Document)
API	Application Programming Interface
QA	Quality Assurance